

# Interview prep sheet

Customer sales data: cleaning and EDA project

Mohammed Naveed | [github.com/NaveedSultan](https://github.com/NaveedSultan)

## How to use this sheet

- Part 1 is the 60-second pitch and the numbers you must know.
- Part 2 has four STAR stories from this project.
- Part 3 explains each piece of code: why you used it and what it changed.
- Part 4 has interview questions with answers you can say out loud.
- Part 5 lists the weak spots in your notebook and report. Read it first. Several answers in Part 4 depend on it.
- Every number here comes from your notebook and report. If a number is not in them, it is not in this sheet.
- This sheet covers the customer sales project only. The churn and AdventureWorks projects need the same treatment.

## Part 1. The pitch and the numbers

### 60-second pitch

I took a raw customer sales file with 10,400 rows and 10 columns. It had duplicate rows, missing values, ages typed as text like "46 years", gender and city written in many different ways, dates stored as text, and 276 rows where the last purchase came before the signup date. I cleaned it in Pandas and ended with 10,197 rows. Then I looked at purchase amount by city, age group, gender and month. Total purchase amount is 408.6 million and the average is about 42,750 per record. Four cities have lower totals, but the reason is fewer records, not lower spending. The biggest finding was about the data itself. About 41% of Customer\_IDs belong to more than one name, so I did not report unique customers, repeat customers or churn.

### Numbers to know

- Raw file: 10,400 rows, 10 columns. After cleaning: 10,197 rows.
- 203 duplicate rows removed. 276 impossible dates flagged. 1,453 age values had text in them.
- Blanks left on purpose: Age 743, Last\_Purchase\_Date 861, Purchase\_Amount 640, Feedback\_Score 504.
- Gender after cleaning: Female 4,856, Male 4,855, Unknown 486.
- Total purchase amount 408,560,586.60. Average 42,749.88. Lowest 5,009.30, highest 79,997.70.
- Eight cities have 33 to 37 million each. Surat, Nagpur, Ahmedabad and Lucknow have 22.6 to 24.5 million.
- Those four cities have 574 to 612 records. The others have 818 to 941. Average per record is about 41,600 to 43,900 everywhere.
- By last purchase year: 2023 121.1M, 2024 125.8M, 2025 126.4M. Growth +3.86% in 2024 and +0.49% in 2025.

- Feedback (9,693 scores): Detractors 40.2%, Passives 30.3%, Promoters 29.5%. Custom bands, not standard NPS.
- Correlation of Age with amount: about -0.01. Feedback with amount: about -0.01.
- Customer\_ID: 2,542 of 6,204 IDs (about 41%) have more than one name. One ID has 6 names.

## Part 2. STAR stories

Keep each story to 60 to 90 seconds. Say the Situation and Task fast, spend most of the time on Action, and finish with numbers in the Result.

### Story 1: Cleaning a messy sales file

#### Situation

I had a raw customer sales CSV with 10,400 rows. It had duplicates, blanks in 8 of the 10 columns, ages stored as text, gender and city typed in different ways, dates stored as text, and 276 rows where the last purchase date came before the signup date.

#### Task

Make the file reliable enough to answer basic questions by city, age group, gender and month, without inventing any values.

#### Action

- Checked the structure first with shape, info and null counts, so I had baseline numbers.
- Pulled the number out of the Age text with a regular expression.
- Converted both date columns to real dates.
- Cleaned Gender and City text so each value had one spelling.
- Removed duplicates after the text cleaning, so rows that differed only by case or spaces were caught.
- Filled missing Name, City, Gender and Email with Unknown. Left missing Age, Purchase\_Amount, Feedback\_Score and Last\_Purchase\_Date blank on purpose.
- Ran checks on emails, negative amounts, feedback range and outliers.

#### Result

10,400 rows became 10,197. Gender has three clean values and City has 12 clean names. Age and dates are real numbers and dates. Every blank that remains is documented, so no made-up value sits inside any total or average.

### Story 2: Finding that Customer\_ID cannot be trusted

#### Situation

I wanted to count unique customers and look at repeat buyers. Both depend on Customer\_ID identifying one person.

#### Task

Check that assumption before building any customer-level analysis on it.

#### Action

- Grouped the data by Customer\_ID and counted how many different names each ID had.
- Left out rows where the name is Unknown, so blanks did not inflate the count.
- Opened individual IDs to see the rows. CUST10004 is Mark Thomas in Pune (signed up 6 Aug 2020) and also Emily Adams in Ahmedabad (signed up 13 May 2021).

#### Result

2,542 of 6,204 IDs (about 41%) have more than one name, and one ID has 6 names. I dropped unique customer, repeat customer, retention and churn analysis, and called the rows "records" in the report. I

wrote down that the data owner needs to explain why IDs repeat.

### **Story 3: Why four cities look weak**

#### **Situation**

The city chart split into two groups. Eight cities had 33 to 37 million. Surat, Nagpur, Ahmedabad and Lucknow had 22.6 to 24.5 million.

#### **Task**

Find out whether those four cities spend less, or just have fewer records.

#### **Action**

- Compared the number of records in each city.
- Compared the average purchase amount per record in each city.

#### **Result**

The four cities have 574 to 612 records against 818 to 941 elsewhere. Average per record is about 41,600 to 43,900 in every city. So the gap comes from volume, not from lower spending. I did not claim to know why those cities have fewer records.

### **Story 4: Handling 276 impossible dates**

#### **Situation**

276 rows had a last purchase date earlier than the signup date. That cannot happen.

#### **Task**

Decide how to treat them without hurting the monthly and yearly numbers and without losing the rows.

#### **Action**

- Created a Date\_Issue column that marks these rows.
- Set their Last\_Purchase\_Date to blank. I could not tell which of the two dates was wrong, so I did not guess.
- Kept the rows and their purchase amounts.
- Ran the date check again and got 0.

#### **Result**

No made-up dates entered the month and year analysis, and the rows still count in totals. The cost: rows with no usable date are left out of the month and year numbers. That is why the yearly amounts add up to 373.3 million while the total is 408.6 million.

## Part 3. The code: why and what it changed

Be ready to explain any line in your notebook in this format: what it does, why you chose it, and what changed because of it.

### 3.1 Load and look before touching anything

```
df = pd.read_csv('...messy_customer_sales_data.csv')
pd.set_option('display.float_format', '{:,.2f}'.format)
df.shape
df.info()
df.isnull().sum()
```

#### Why I used it

- You need a baseline before you change anything.
- info() shows the data types. That is where you see Age and both date columns are text.
- The float format option shows money with commas and two decimals, so totals are easy to read.

#### What it changed

- Baseline: 10,400 rows, 10 columns, and the blank count for every column.
- These numbers let you prove later what the cleaning removed.

### 3.2 Pull the number out of the Age text

```
def get_age(x):
    nums = re.findall(r'\d+', str(x))
    return int(nums[0]) if nums else np.nan

df['Age'] = df['Age'].apply(get_age)
```

#### Why I used it

- Values like "46 years" cannot be averaged. The pattern \d+ finds the digits and the function takes the first group.
- str(x) is there for blank values. A blank becomes the text 'nan', has no digits, and returns NaN instead of crashing.

#### What it changed

- 1,453 text values turned into numbers.
- Age is now numeric, so averages, age groups, scatter plots and correlation all work.

### 3.3 Convert dates

```
df['Signup_Date'] = pd.to_datetime(df['Signup_Date'], errors='coerce')
df['Last_Purchase_Date'] = pd.to_datetime(df['Last_Purchase_Date'], errors='coerce')
```

#### Why I used it

- Text dates cannot be subtracted or grouped by month.
- errors='coerce' turns a date it cannot read into NaT (a blank date) instead of stopping the code.

#### What it changed

- Both columns are real dates.
- This made the impossible-date check, the month grouping and the days-between KPI possible.

### 3.4 Standardize Gender

```
df['Gender'] = df['Gender'].str.strip().str.lower()
df['Gender'] = df['Gender'].replace(
    {'m':'Male', 'male':'Male', 'f':'Female', 'female':'Female'})
```

#### Why I used it

- strip() removes extra spaces and lower() removes case differences, so the mapping needs fewer keys.
- replace() maps each spelling to one label.

#### What it changed

- Gender ends with Female, Male and Unknown only.
- Warning: replace() leaves any spelling it does not know as it is. Check value\_counts() to confirm nothing is left over.

### 3.5 Standardize City

```
df['City'] = df['City'].str.strip().str.title()
```

#### Why I used it

- 'JAIPUR', 'jaipur' and 'Jaipur' are three different values to Pandas.

#### What it changed

- 12 clean city names. Without this, a group-by would split one city into several rows and the city totals would be wrong.

### 3.6 Age range guard

```
df.loc[(df['Age'] < 0) | (df['Age'] > 100), 'Age'] = np.nan
```

#### Why I used it

- A safety check against impossible ages. Anything outside 0 to 100 becomes blank.

#### What it changed

- No values were found outside the range.
- Honest note: the regex only reads digits, so it can never produce a negative age. Only the above-100 half of this check could ever fire.

### 3.7 Remove duplicates

```
df = df.drop_duplicates().reset_index(drop=True)
```

#### Why I used it

- It runs after the text cleaning on purpose. Rows that only differed by case or spaces now match exactly and get caught.
- By default it compares every column and keeps the first copy.
- reset\_index(drop=True) gives clean row numbers again.

#### What it changed

- 203 rows removed. 10,400 became 10,197.
- Duplicate rows would have counted the same purchase twice in every total.

### 3.8 Fill missing text with Unknown

```
for col in ['Name', 'City', 'Gender', 'Email']:
    df[col] = df[col].fillna('Unknown')
```

#### Why I used it

- Dropping the row would throw away its other columns. A row with no email can still have a valid purchase amount.
- Unknown keeps the row and shows how big the gap is.

#### What it changed

- Gender Unknown is 486 rows. Unknown city rows were left out of the city chart so the chart only shows known locations.

### 3.9 Validation checks

```
emails[~emails['Email'].str.contains(r'@.\.', regex=True)]
df[df['Purchase_Amount'] < 0]
df['Feedback_Score'].min(), df['Feedback_Score'].max()
df[df['Last_Purchase_Date'] < df['Signup_Date']]
```

#### Why I used it

- Each line tests a rule the data must follow: an email has an @ and a dot after it, amounts are not negative, scores stay in range, a last purchase cannot come before signup.

#### What it changed

- Emails: 9,559 checked, none failed. Amounts: none negative, minimum 5,009.30. Feedback: 1 to 10. Dates: 276 rows broke the rule.
- The email test is a basic one. Say 'none failed a basic check', not 'none were malformed'.

### 3.10 Flag and blank the impossible dates

```
df['Date_Issue'] = df['Last_Purchase_Date'] < df['Signup_Date']
df.loc[df['Date_Issue'], 'Last_Purchase_Date'] = pd.NaT
```

#### Why I used it

- I could not know which date was wrong, so I blanked the one used in the time analysis.
- The flag column keeps a record of which rows were touched.

#### What it changed

- 276 rows flagged. The re-check returned 0.
- These rows keep their purchase amount in the totals but are missing from month and year numbers.

### 3.11 Age groups

```
df['Age_Group'] = pd.qcut(df['Age'], q=3, labels=['Young', 'Adult', 'Senior'])
df['Age_Group'] = df['Age_Group'].cat.add_categories('Unknown')
df['Age_Group'] = df['Age_Group'].fillna('Unknown')
```

#### Why I used it

- qcut splits the ages into 3 groups with the same number of people in each.
- The column is a category type. It refuses a new label like Unknown until add\_categories() adds it.

#### What it changed

- Young 18 to 35, Adult 35 to 53, Senior 53 to 70, plus 743 rows labelled Unknown.
- Because qcut forces equal groups, equal group sizes are not a finding. See Part 5.

### 3.12 Outlier check

```
q1, q3 = df[col].quantile([0.25, 0.75])
iqr = q3 - q1
n = ((df[col] < q1 - 1.5*iqr) | (df[col] > q3 + 1.5*iqr)).sum()
```

#### Why I used it

- The box plot shows outliers visually. The IQR rule counts them: any value more than 1.5 times the middle-50% spread beyond the middle 50% is flagged.

#### What it changed

- 0 outliers in Age, Purchase\_Amount and Feedback\_Score, so no rows were removed or capped.

### 3.13 The charts

```
city_data = df[df['City'] != 'Unknown'].groupby('City')['Purchase_Amount'].sum()
df['Gender'].value_counts()
df['Age_Group'].value_counts()
df.groupby(df['Last_Purchase_Date'].dt.to_period('M'))['Purchase_Amount'].sum()
pd.crosstab(df['City'], df['Age_Group'])
df[['Age', 'Purchase_Amount', 'Feedback_Score']].corr()
```

#### Why I used it

- Group-by with sum gives total amount per city. value\_counts gives record counts for gender and age group.
- to\_period('M') groups dates by month. crosstab counts records for each city and age group pair.
- corr() with a heatmap shows which numbers move together.

#### What it changed

- Showed the two city groups, the equal gender split, and a monthly line with no clear direction.
- Showed that age, amount and feedback have almost no link (about -0.01).

### 3.14 Customer\_ID check

```
df.groupby('Customer_ID')['Name'].nunique().sort_values(ascending=False)
```

#### Why I used it

- If an ID identifies one customer, each ID should have exactly one name. This line counts the names per ID.

#### What it changed

- 41% of IDs fail. That stopped me from reporting unique customers, repeat buyers, retention or churn, all of which would have been wrong.
- Check: this cell counts Unknown as a name. Your report says Unknown rows were ignored. See Part 5.

### 3.15 KPIs

```
df['Purchase_Amount'].agg(['sum', 'mean', 'max', 'min'])
df['Days_To_Last_Purchase'] = (df['Last_Purchase_Date'] - df['Signup_Date']).dt.days
monthly_Sales.pct_change() * 100
yearly_Sales.pct_change() * 100
```

#### Why I used it

- agg() gives four numbers in one line. pct\_change() compares each period with the one before it.
- Pandas skips blanks in sum and mean, so blank amounts do not break them.

#### What it changed

- Total 408,560,586.60, average 42,749.88 (calculated on 9,557 rows that have an amount).
- Growth: 2024 +3.86%, 2025 +0.49%. Month on month swings between about -21% and +27%.

### 3.16 Feedback bands

```
df['Feedback'] = np.where(df['Feedback_Score'] >= 8, 'Promoter',
    np.where((df['Feedback_Score'] >= 5) & (df['Feedback_Score'] < 8), 'Passives',
    np.where((df['Feedback_Score'] >= 1) & (df['Feedback_Score'] < 5), 'Detractors',
    'No_Feedback'))
```

#### Why I used it

- Nested np.where checks the score band by band.
- A blank score fails every comparison, so it falls through to No\_Feedback. That is why the last label works.

#### What it changed

- Promoters 2,858, Passives 2,939, Detractors 3,896, No feedback 504.
- Detractors are 40.2% of the 9,693 scores. These are custom bands, not standard NPS.

## Part 4. Interview questions and answers

Answers are written the way you would say them. Change the wording to sound like you, but keep the facts. If you do not know something, say so. That is better than guessing.

### A. The project

#### Q1. Tell me about this project.

Use the 60-second pitch from Part 1.

#### Q2. Where did the data come from?

It is a practice file called `messy_customer_sales_data.csv`. I used it to show how I clean and explore data, so I treat the results as what this file says. I do not claim they describe a real company.

If you got the file from a named public source, say the name. Do not hide that the data may be generated. See Part 5, item 9.

#### Q3. What questions were you trying to answer?

Which cities bring in the most purchase amount, whether age group or gender changes spending, how purchase amount moves by month, and what the feedback split looks like. I also wanted basic KPIs such as total, average and growth.

#### Q4. Why Python and not Excel or SQL?

At 10,000 rows any of them works. I used Python because the regex for Age, the checks and the charts sit in one notebook that I can rerun from the top. In SQL I would use TRIM, LOWER and CASE WHEN for the text, ROW\_NUMBER to find duplicates, and GROUP BY for totals.

### B. Cleaning decisions

#### Q5. Walk me through your cleaning steps and why in that order.

First I looked at the data: shape, types and blank counts. Then I fixed types, so Age became numbers and the dates became dates. Then I fixed the text in Gender and City. Then I removed duplicates, because only after the text was clean did near-copies become exact copies. Then I handled blanks, ran checks, and added the flag and group columns.

#### Q6. Why a regular expression for Age?

Some ages looked like "46 years". The pattern `\d+` finds the digits so I can keep the number.

Follow-up you may get: can you do it without apply? Yes, `df['Age'].astype(str).str.extract(r'(\d+)')` does the same and is faster on a whole column.

#### Q7. What does `errors='coerce'` do, and what is the risk?

It turns a date Pandas cannot read into NaT instead of raising an error. The risk is that bad dates become blank without a warning, so I count blanks after the conversion to see what changed.

#### Q8. Why remove duplicates after standardizing the text?

Before cleaning, a row with "Jaipur" and a copy with "JAIPUR" look different. After cleaning they match, and `drop_duplicates` catches them. Doing it in the other order would leave some copies behind.

**Q9. Why did you not fill missing Age or Purchase\_Amount with the average?**

743 ages and 640 amounts are blank. If I filled them with the average, hundreds of rows would carry a value I made up, and the total of 408.6 million would include numbers that are not in the data. Pandas skips blanks in averages, and the Unknown age group shows how many are missing.

**Q10. Why Unknown for text columns instead of dropping the rows?**

A row with no email can still have a good purchase amount. Dropping it loses that. Unknown keeps the row and shows how big the gap is. For the city chart I left Unknown out, because it is not a real location.

**Q11. What did you do about the 276 impossible dates, and why?**

I flagged them in a Date\_Issue column and set the last purchase date to blank. I could not tell whether the signup date or the purchase date was wrong, so I did not guess. I kept the rows because their amounts are still valid for city and age analysis.

Trade-off to admit: those rows are missing from month and year numbers.

**Q12. How did you check your cleaning worked?**

I compared shape before and after (10,400 to 10,197). I ran value\_counts on Gender and City, ran isnull().sum() again, and re-ran the date check, which returned 0.

**Q13. How did you choose the age groups?**

I used qcut with three groups, which gives about the same number of people in each: 18 to 35, 35 to 53 and 53 to 70. Because qcut forces equal sizes, I do not treat the equal sizes as a result. If I did it again I would use fixed bands people recognise, such as 18 to 30, 31 to 45, 46 to 60 and over 60.

**Q14. You found no outliers anywhere. Does that look right?**

Purchase amounts run from 5,009 to 79,998 and feedback runs from 1 to 10, with nothing far from the rest. The IQR rule found 0 outliers in all three columns. Real sales data usually has a few very large purchases, so this also tells me the file is probably generated. I did not drop or cap anything.

**C. Analysis and results**

**Q15. What did you find by city?**

Eight cities have 33 to 37 million each, with Chennai, Mumbai and Pune at about 37 million. Surat, Nagpur, Ahmedabad and Lucknow have 22.6 to 24.5 million.

**Q16. Does that mean those four cities are weak markets?**

No. A city total is the number of records times the average amount. Those four cities have 574 to 612 records against 818 to 941 elsewhere, and the average per record is about the same everywhere. So volume differs. The data does not tell me why, and I would ask the data owner.

**Q17. What did you find for age and gender?**

Female and male records are almost equal (4,856 and 4,855). Median purchase is about 42,800 to 43,600 across age groups, and the correlation between age and amount is about -0.01.

If asked about a significance test: I did not run one. Correlation only checks for a straight-line link. A test for differences between groups would be the next step.

**Q18. What does the monthly trend show?**

Amount moves between 8.4 million (Apr 2023) and 11.9 million (Oct 2023) with no clear direction. Each row has one amount and one last purchase date, so this chart shows when customers last bought. It is not real monthly sales.

**Q19. Growth was 3.86% in 2024 and 0.49% in 2025. Is the business slowing down?**

I cannot say that. These numbers use last purchase date, and each customer row has only one date, so they are not sales growth. Also, about 35 million of the total amount has no usable date and is not in these year totals. I would call it a rough signal at most.

**Q20. Why did you use custom feedback bands?**

I split the 1 to 10 score into 8 to 10, 5 to 7 and 1 to 4. These are my own bands and I labelled them that way in the report. Standard NPS uses a 0 to 10 scale with 9 to 10 as promoters, 7 to 8 as passives and 0 to 6 as detractors, so I did not call my result NPS.

**Q21. 40% of scores are in the lowest group, yet those records spend as much as others. What do you make of that?**

In this file, feedback and amount have a correlation of about -0.01, so unhappy records spend about as much as happy ones. I would not say feedback does not matter. Each row has only one amount, so I cannot see whether unhappy customers stopped buying later.

**Q22. Your total is 408.6 million but the yearly amounts add up to 373.3 million. Why?**

861 rows have no usable last purchase date. Some were blank in the raw file and 276 I blanked myself. Their amounts count in the total but cannot be placed in a year. That is about 35 million, roughly 9% of the total.

**Q23. What does the average of 1,318 days from signup to last purchase tell you?**

Not much. Signups run from 2018 to 2023 and last purchases from 2023 to 2025, so the number mostly reflects those date ranges. I included it as a KPI but I would not base a decision on it.

**D. Customer\_ID****Q24. How is the Customer\_ID problem different from the duplicates you removed?**

Duplicates are rows that match in every column, and I removed 203 of them. Here the same ID belongs to different people with different names, ages and cities. I cannot remove those rows, because they are different records.

**Q25. Why not just fix the IDs?**

I cannot tell which rows belong to the same real person. Guessing would add errors. The data owner has to explain whether IDs are reused or mixed up.

**Q26. What would you do with a real company's data?**

Ask the data owner or check the source system to find out how IDs are created. I would test whether Email works better as a key before using it, since 655 emails are missing in this file. Until then I would report records, not customers.

**E. Technical questions**

**Q27. replace() versus map() for Gender?**

replace() leaves values it does not know as they are. map() turns unknown values into blanks. I used replace, so I checked value\_counts() afterwards to make sure no old spelling remained.

**Q28. What is the difference between NaN and NaT?**

NaN is a missing number. NaT is a missing date. Both are skipped in sums and averages.

**Q29. qcut versus cut?**

cut makes groups from ranges you choose, such as 18 to 30. qcut makes groups with equal counts of rows.

**Q30. How does the IQR rule work?**

Find the 25th and 75th percentile. The gap between them is the IQR. A value more than 1.5 times the IQR below the 25th or above the 75th percentile is an outlier.

**Q31. What does pct\_change do?**

It compares each value with the one before it: (this - previous) / previous. The first row is blank.

**Q32. How would you get the city totals in SQL?**

```
SELECT City, SUM(Purchase_Amount) FROM sales WHERE City <> 'Unknown' GROUP BY City  
ORDER BY 2 DESC;
```

**F. Behavioral questions****Q33. What would you do differently?**

Use fixed age bands instead of qcut. Make sure every number in the report has a cell in the notebook that produces it. Ask earlier where the data came from, and test Customer\_ID before planning any customer-level analysis.

**Q34. What was the hardest part?**

Pick the true one for you. From your notebook, the honest choice is the Customer\_ID finding, because it meant dropping analysis I had planned and explaining why.

**Q35. Explain the results to a manager with no technical background.**

Spending per record is about the same in every city, age group and gender. Four cities have lower totals because we have fewer records there, not because people spend less. About 4 in 10 feedback scores are in the lowest group, but they do not link to how much people spend. We cannot count unique customers yet because the same ID is used for different people, so that needs fixing first.

## Part 5. Fix these before the interview

These are real gaps between what your report says and what your notebook shows. An interviewer who opens the notebook can find any of them. Fixing them takes little time.

### 1. The equal age groups are not a finding.

Your report says the three age groups are close in size. `qcut` builds equal groups on purpose. Remove that line from the chart caption, or rebuild the groups with fixed bands.

### 2. The negative-age check can never fire.

The regex reads digits only, so it cannot return a negative number. Say that if asked, or drop the below-zero half of the check.

### 3. Numbers in the report that no notebook cell produces.

The average amount per record by city, the median purchase by age group, and the `Customer_ID` count that ignores Unknown names are in the report but not in the notebook. Cell 67 counts Unknown as a name. Add cells for all three, so you can show them live.

### 4. Notebook charts do not match the report.

The notebook city chart is in thousands and has a typo ("Thounsands"). The report shows millions. The monthly chart title in the code still says "Monthly sales" while the report says it is not real monthly sales. Gender and age charts say "customers", the report says "records". Make them match.

### 5. The duplicate count is described loosely.

The report says 203 exact duplicate rows in the raw data. The notebook removes duplicates after the text cleaning and never counts them in the raw file. Either run the count on the raw file, or change the wording to "203 duplicate rows after standardizing".

### 6. Yearly totals do not add up to the total.

373.3 million by year against 408.6 million overall. About 35 million (roughly 9%) belongs to rows with no usable date. Put one line about this in the report.

### 7. The email check is weak.

The pattern `@.+\\.` accepts many bad emails. Say "none failed a basic check", not "none were malformed".

### 8. Do not call the feedback split NPS.

Your bands are custom (8 to 10, 5 to 7, 1 to 4). Standard NPS uses 0 to 10 with 9 to 10, 7 to 8 and 0 to 6. You already say this in the report. Keep saying it out loud.

### 9. The data looks generated.

Correlations near zero, an almost exact male and female split, amounts spread evenly from 5,009 to 79,998 and no outliers in any column all point that way. If you say so first, it shows judgment. If the interviewer says it first and you did not notice, it hurts you.

### 10. Check the Gender mapping.

Your report lists "FEM" as a raw spelling, but the mapping in cell 20 has no key for it. Since only three values remain, the spelling may not exist. Run `value_counts()` on the raw column to confirm, and add any missing spelling to the dictionary.

### **11. Hard-coded file path and repeated code.**

The path starts with D:\, which breaks for anyone else. Use a relative path such as data/messy\_customer\_sales\_data.csv. The qcut line also runs twice (cells 37 and 38).